

Danuja Wijerathna

AI Engineer | AI Solution Architect

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PROFESSIONAL SUMMARY

AI Engineer with 3+ years building production AI systems on Microsoft Azure. Day-to-day work covers multi-agent orchestration on Azure AI Foundry, Agentic RAG over enterprise data, MCP-based tool integration, and the LLMOps pipelines that ship these systems safely. Comfortable end-to-end: Azure Data Factory ingestion, Microsoft Fabric semantic layers, Azure AI Search hybrid retrieval, knowledge graphs, and Azure DevOps CI/CD. Increasingly stepping into solution-architect work: target architectures, build-vs-buy decisions, and design reviews on cross-team AI initiatives.

PROFESSIONAL EXPERIENCE

Senior AI Engineer · IFS R&D International (Pvt) Ltd · Colombo, Sri Lanka

Oct 2025 – Present

- Architected multi-agent AI systems on **Azure AI Foundry** using **Microsoft Agent Framework**, with specialised agents handling HR, finance, and customer-support workflows.
- Built **Agentic RAG** pipelines that combine **Graph RAG**, hybrid vector and semantic search on **Azure AI Search**, and conversation-memory management for multi-turn reasoning over enterprise data.
- Built **Model Context Protocol (MCP)** servers that expose ERP, ticketing, and HR systems as governed tools for AI agents, with audit-ready traces on every action.
- Defined the **LLMOps** reference architecture across multi-subscription Azure environments: Terraform and Bicep for IaC, automated CI/CD pipelines, model evaluation, observability, and Responsible AI guardrails for production Generative AI workloads.
- Built ingestion and ETL pipelines on Azure Data Factory and Azure Databricks that feed vector stores and Lakehouse layers with curated, lineage-tracked data from structured and unstructured sources.
- Built reusable semantic layers in Microsoft Fabric over large enterprise datasets so AI agents and BI consumers query consistent business metrics through one model.
- Shipped AI-powered developer tools (automated code analyser and test-script generator), cutting QA review effort by about 25 hours per week.
- Built time-series forecasting models for employee utilisation at 87% accuracy, used for resource planning across delivery teams.
- Acted as solution architect on cross-team AI initiatives: target architectures, build-vs-buy decisions, and design reviews with engineering leadership.

Senior Data Scientist · IFS R&D International (Pvt) Ltd · Colombo, Sri Lanka

Sep 2024 – Sep 2025

- Led migration of legacy Oracle databases to a modern Azure data architecture (Azure Data Factory, Azure Synapse Analytics, ADLS Gen2), with better query performance and scalability.
- Implemented a GDPR-compliant, AI-driven employee performance tracking system with row-level security using Power Apps, Power BI, and Dataverse.
- Designed and automated end-to-end ETL/ELT pipelines in Azure Data Factory and Databricks for cross-system data integration, warehousing into Synapse and Microsoft Fabric.

Data Scientist · IFS R&D International (Pvt) Ltd · Colombo, Sri Lanka

Sep 2023 – Sep 2024

- Designed semantic models in Microsoft Fabric for large enterprise datasets, which sped up Power BI reports across business units.
- Built a customer churn model using deep learning (Artificial Neural Networks) at 89% accuracy, used by account-management teams to prioritise retention.

Data Engineer · AI Palette · Singapore

Dec 2022 – Aug 2023

- Built real-time data processing pipelines and cleaned up existing ETL workflows, which cut time-to-insight for product analytics teams.
- Designed an analytics layer over the data lake that supported custom analysis and consumer-behaviour insights for downstream ML and BI consumers.
- Built ETL solutions for loading CSV and Parquet files into data lakes and cloud warehouses, used by BI and machine learning teams.

Data Science Intern (Undergraduate Trainee) · IFS R&D International (Pvt) Ltd · Colombo, Sri Lanka

Dec 2021 – Dec 2022

- Automated the Global Customer Support process, which reduced manual workload and improved response times across multi-region teams.
- Built a customer-solution prediction model in TensorFlow that anticipated customer needs and recommended proactive support actions.

KEY PROJECTS

ServiceNow MCP Server: Agentic ITSM Intelligence Platform

Stack: Model Context Protocol (MCP), ServiceNow REST APIs, Graph Database (multi-hop traversal), Azure AI Foundry IQ (multi-index retrieval), K-Means Clustering, Community Detection, Graph Deep Learning, NLP, Agentic AI

- Designed and built a **Model Context Protocol (MCP) server** that exposes ServiceNow as a governed toolset for AI agents: **record queries** across ServiceNow tables, **full incident lifecycle management** (create, update, triage, resolve), and **multi-step ITSM workflow automation** from a single conversational request — pluggable into any MCP-compliant agent framework with governed, auditable tool access.
- Modeled the ServiceNow estate as a **knowledge graph in a graph database** — incidents, cases, customer interactions, and their relationships — so agents reason over connected records through **multi-hop traversal** instead of isolated table lookups, surfacing links between tickets that flat queries can never see.
- Unified **RAG grounding** through **Azure AI Foundry IQ as a multi-index wrapper**: multiple Azure AI Search indexes (incidents, knowledge articles, related records) registered as knowledge sources behind one governed, MCP-compatible retrieval endpoint, with agentic query planning and **semantic ranking** federated across all sources instead of per-index integration plumbing.
- Built graph-powered **root cause analysis (RCA)**: multi-hop traversal walks the relationship chains behind an incident to trace symptoms back to their origin, producing RCAs grounded in the actual chain of connected records rather than single-ticket summaries.
- Added unsupervised **pattern discovery** over the incident corpus: **K-Means clustering** groups incidents by similarity to expose recurring problem classes, while **community detection** over the interaction graph reveals dense clusters of related records — turning thousands of individually-handled tickets into a ranked map of systemic issues.
- Built a **graph-based deep sentiment model** on the same graph: instead of scoring a single text field, it maps the full customer interaction graph across ticket history to pinpoint the customer's **actual sentiment along with its root cause**, giving support teams a reason to act, not just a score.

AI Power BI Report Builder: AI-Assisted Reporting Co-Working Space

Stack: Electron, React, TypeScript, Tailwind, Azure AI Foundry (GPT-5.4), Anthropic Claude, Model Context Protocol (MCP), Ajv, electron-builder (NSIS)

- Sole designer and developer of a Windows desktop app (Electron, React, TypeScript) that lets users build Power BI reports through natural-language chat. Built a multi-agent system where an Orchestrator classifies user intent and routes to one of six specialist agents (Analysis, Model, Visual, DAX, Repair, Fallback), each with its own system prompt, tool whitelist, and bounded agentic loop.
- Designed a Report Tree Index that compresses a `.pbip` project into a token-efficient structural summary, cutting LLM token usage on visual-edit tasks by about 95%, with lazy on-demand expansion through an MCP tool.
- Built a safe-write pipeline that prevents Power BI Desktop from crashing on generated files: a Data Gate strips customer values before any prompt leaves the machine, Ajv validates JSON against six Microsoft PBIR schemas,

six CrashPrevention rules auto-fix known gotchas, and a backup-then-atomic SafeWriter rolls back any failed edit.

- Built a bilingual LLM client that speaks both Azure AI Foundry (Azure OpenAI Chat Completions) and Anthropic Messages API with auto-detection from endpoint URL, plus real-time SSE streaming forwarded from the Electron main process to the renderer over IPC.
- Connects to a Power BI Modeling MCP server for semantic-model and DAX operations. Added a DreamService that consolidates chat history into long-lived memory at conversation reset. Ships as a single-click Windows installer.

Project Helix: Conversational BI for Enterprise Leadership

Stack: Python 3.11, FastAPI, Claude Agent SDK (Claude Opus 4.8), Azure AI Foundry (fallback), in-process MCP tools, Power BI / Microsoft Fabric, live DAX, MSAL, DuckDB, PyArrow, React, Server-Sent Events

- Designed and built a multi-agent conversational-BI platform ("Ask Helix") that lets enterprise leadership query certified Power BI semantic models in natural language. A Supervisor agent classifies intent (LLM-first, with a regex offline fallback) and routes to domain specialists; each specialist composes three sub-agents (live DAX data, ML insights, business-glossary context) behind a single assistant voice, streamed token-by-token over Server-Sent Events with inline charts and tool-call traces.
- Removed the need for leadership to comb through each Power BI report by hand: a Helix supervisory agent connects to and monitors every report, surfaces potential issues, and produces a root-cause analysis alongside them, so reviewers can start investigating right beside the live report the agent is listening to.
- Migrated the LLM backend from Azure AI Foundry to the Claude Agent SDK (Claude Opus 4.8), exposing seven read-only tools as in-process MCP tools and locking out all built-in tools for governance-by-construction, with the original engine kept as a hot-swappable fallback. Built DataGate, a PII-masking firewall between tool execution and the model, so employee-level data never reaches the LLM while authorised users still see full results.
- Integrated live Power BI / Fabric via MSAL with silent token refresh and validated DAX accuracy against certified measures (matched the published report to the decimal). Added engine-aware cost and context-headroom telemetry, daily-rotated JSONL audit logging, and a bounded agentic loop.

Virtual Resource Manager

Stack: FastAPI (Python 3.11), React 18, TypeScript, Azure AI Foundry (GPT-5.4-nano, GPT-5.2, text-embedding-3-large, Claude Opus 4.6 fallback), Azure AI Search, Azure SQL, MSAL / Azure AD, Azure Container Apps, Logic Apps

- Sole AI engineer and architect of an enterprise-grade multi-agent AI assistant that matches employees to incoming project requests across 854 profiles. Built an Orchestrator plus five specialist sub-agents (Skills, Availability, Historical Project, Ranking, LLM Query Parser) with parallel execution and streaming SSE response synthesis. Embedded inside the company's internal AI platform with Azure-AD-resolved RBAC headers injected by an Express gateway.
- Designed a 10-dimensional configurable scoring engine (Skill Match, Availability, Customer Experience, Industry Experience, plus six others) with DB-backed weight profiles shared between the Resource Manager Search and Chat surfaces, holding score parity to within 0 to 3% across both.
- Built the hybrid Azure AI Search layer on 854 employee profiles: BM25 keyword search plus HNSW vector search over two 3072-dimensional embeddings, semantic re-ranking, and a custom scoring profile boosting skills 5x and offering 3x. Added a three-tier skill-enrichment chain (hybrid search over historical JIRA tickets, curated SQL mappings, then LLM extraction) with a 500-entry LRU at 15-minute TTL and a 3-second hard timeout so slow lookups never block the request path.
- Hot-path performance wins: cut a 39-second skill-lookup to zero by reading directly from AI Search rather than SQL, collapsed 126 individual SQL roundtrips into one batched call, and added a separate light connection pool so polling endpoints cannot starve the heavy pipeline. Multi-model LLM strategy across Azure AI Foundry (GPT-5.4-nano on the hot path, GPT-5.2 fallback, Claude Opus 4.6 as ultimate fallback) with full per-model cost tracking.

Log Analyzer: AI Root Cause Analysis Platform

Stack: FastAPI (Python 3.11), PostgreSQL (JSONB), Alembic, Parquet, DuckDB, Azure OpenAI (GPT-5.2), Microsoft Agent Framework, Azure AI Search, Azure Container Apps, Server-Sent Events, React 18

- Sole architect and developer of an AI-powered Root Cause Analysis platform for enterprise Mobile Work Order incidents. Ingests heterogeneous logs (MAUI client traces, Android `bugreport` dumpstates, iOS `.ips` crash files, cloud server sync traces) at tens of millions of lines per case, and runs an 11-agent AI pipeline across six stages that produces an executive-grade RCA with every claim cited back to the underlying log shards.

- Built a Parquet-shard ingestion pipeline that normalises four log formats into a 30-second-windowed common schema readable by DuckDB. Solved the small-files problem with an `aiolimiter`-throttled, `ThreadPoolExecutor`-parallel six-step ladder, achieving order-of-magnitude performance wins without changing storage backend.
- Led the full migration from blob-first persistence to PostgreSQL JSONB as single source of truth: six blob writes per RCA collapsed to two, warm-read latency on analytics tabs cut from 5 to 94 seconds down to sub-second, and error-mapping cold-to-warm from 55 seconds to 270 milliseconds. Shipped via additive Alembic migrations with feature-flagged progressive deployment and 422 passing tests.
- Real-time SSE pipeline with a bounded 200-event replay buffer, so users can switch tabs mid-RCA and resume cleanly. Per-finding enrichment via Azure AI Search across JIRA defects (with scoring profiles and semantic configs), ServiceNow incidents, and a 23,213-document Oracle error taxonomy that classifies findings and maps them to known resolutions.

TECHNICAL SKILLS

AI & Generative AI: Large Language Models (LLMs), Agentic AI, Multi-Agent Orchestration, Retrieval-Augmented Generation (RAG), Graph RAG, Agentic RAG, Knowledge Graphs, Vector Embeddings, Semantic Search, Hybrid Search, Prompt Engineering, Fine-tuning, Model Context Protocol (MCP)

Azure AI Platform: Azure AI Foundry, Foundry Agent Service, Azure OpenAI Service, Azure AI Search, Azure Document Intelligence, Azure Cognitive Services, Azure Machine Learning

Agent Frameworks: Microsoft Agent Framework, Claude Agent SDK, LangChain, LangGraph, Semantic Kernel, AutoGen

Data Platform: Azure Data Factory (ADF), Azure Databricks, Microsoft Fabric, Azure Synapse Analytics, Azure Data Lake (ADLS Gen2), Lakehouse Architecture, Semantic Models, Power BI, Power Apps, Dataverse, Delta Lake, PostgreSQL, DuckDB

MLOps / LLMops: Azure DevOps, GitHub Actions, MLflow, Azure ML Prompt Flow, Terraform, Bicep, Docker, CI/CD Pipelines, Model Evaluation & Monitoring, Drift Detection, Infrastructure as Code (IaC)

Programming: Python, TypeScript, JavaScript, SQL, PySpark

Architecture & Cloud: Solution Architecture, Cloud-Native Design, Microservices, Serverless, REST APIs, Microsoft Entra ID, Zero Trust, Responsible AI Governance, GDPR Compliance, Row-Level Security

Machine Learning: Deep Learning, Artificial Neural Networks (ANN), Time Series Forecasting, Natural Language Processing (NLP), TensorFlow, PyTorch, scikit-learn, Pandas, NumPy

EDUCATION

MSc in Applied Artificial Intelligence — Jan 2026 – Present

University of Westminster, United Kingdom

BSc (Hons) in Information Technology, Specialisation: Data Science — 2019 – 2023

Sri Lanka Institute of Information Technology (SLIIT), Malabe

ACHIEVEMENTS

- IFS R&D High-Performance Individual recognition (2025): selected for the company's High-Performance Team.
- IFS Aspire Leadership Programme (2025): nominated for performance-based leadership development.
- NBQSA National ICT Award (2023): Finalist, Top 5 Inter-University in Sri Lanka.
- SLASSCOM National Ingenuity Awards (2022): Best Innovative Regional Project.
- STEMUP Hack AI (2022): Finalist, Top 5 Inter-University in Sri Lanka.
- Dean's List Award (2021).
- Google Kick Start & Hash Code (2021): global rankings 13,429 and 6,661 respectively.